Gampa Sai Sasivardhan

Roll No.: 22CHB0B13
B.Tech 3rd Year
Chemical Engineering
National Institute Of Technology, Warangal

+91-7661991859
gampassv2004@gmail.com
gs22chb0b13@student.nitw.ac.in
GitHub Profile
LinkedIn Profile

Education

•National Institute of Technology Warangal *2023-Present*
B.Tech in Electronic and Communication Engineering CGPA: 9.00

•National Institute of Technology Warangal *2022-Present*
B.Tech in Chemical Engineering CGPA: 7.58

•Narayana College *2020 - 2022*
Board of Intermediate Education, Telengana Percentage: 95.8%

•Pallavi Model School *2009 - 2020*
Central Board of Secondary Education Percentage: 92.8%

Experience

•Mowito Robotics *Jun 2024 - Jul 2024*
Project Intern Bengaluru , India

- Developed an eccentricity checking machine, achieving 97% accuracy, involving precision engineering and ROS2 integration with STM32 using OpenCV.
- Designed a robotic conveyor system, increasing automation efficiency by 40%, utilizing Fusion360 and STM32 integration.

•Robotics Club NITW *Jan 2023 - Present*
Project Incharge Warangal , Telengana , India

- Developed advanced robotic arms and AGVs, improving operational efficiency by 25%, using ROS2 and precision engineering techniques.
- Led the team in eYantra and Robocon, showcasing innovative robotic solutions, focusing on autonomous navigation and object manipulation.

Personal Projects

•Moon Rover *Jan 2024 - Present*
An ISRO's robotic competition called IRoC-U.

- Developed and executed an advanced autonomous rover solution with auto-navigation and robotic arm capabilities, increasing operational efficiency by 40%, using V-SLAM and PID algorithms.
- Developing a V-SLAM navigation model, improving accuracy by 30% using Intel Real Sense Depth Camera on Nvidia Jetson Nano for image processing.
- Implemented PID algorithm for 30% smoother movement of robotic arm and speed control of rover motion in sandy terrains using Raspberry Pi, achieving precise object picking with Inverse Kinematics.

•Waste Management Robot (WMR) *Dec 2023*
Efficient waste management robot using Arduino IDE for sensor input and 3-axis robotic arm for segregation.

- Designed a sensor pad with 85% accuracy for identifying waste like plastic, metal, paper, wet etc., using sensors and Arduino Mega.
- Developed a 3-axis robotic arm, increasing segregation efficiency by 45%, for sorting waste into respective dustbins.
- Engineered an omnidirectional rover with a 15 kg holding capacity using Arduino Uno, improving waste handling efficiency by 45%.

•5-Axis Robotic Arm *Jan 2024*
Advanced 5-axis robotic arm employs OpenCV for color recognition and inverse kinematics for precise object placement

- Created a 5-axis robotic arm using high torque servo motors, achieving 95% precision in segregation applications.
- Leveraged advanced OpenCV color recognition to achieve 90% accuracy in object identification, employing inverse kinematics for precise placement.
- Executed PID algorithm for 30% smoother servo movement, ensuring precise control with minimal error, enhancing overall performance.

•GPS-GSM Coordinate Transmission System *Jul 2023*

Integrated GSM and GPS modules with Arduino Nano for live location

- Engineered a solution for real-time GPS coordinate transmission via SMS, reducing response time by 30%.
- Implemented SoftwareSerial library for seamless communication between Arduino board and GSM/GPS modules, achieving 95% data accuracy with TinyGPS++ library.
- Architected a system for instantaneous GPS coordinate sharing via SMS, enhancing emergency response efficiency by 40%.

•RFID Attendance System *Nov 2023*

Designed an RFID attendance system to automate and streamline attendance tracking.

- Designed and implemented an RFID attendance system, automating student attendance tracking and increasing accuracy by 25%.
- Developed an interface to upload attendance data to Google Sheets, reducing manual errors by using Google Apps Script.
- Enhanced accuracy and efficiency in attendance management, providing real-time data access and reducing manual errors by 95%.

Technical Skills and Interests

Languages: Python , Javascript , C++ , Embedded C, HTML , CSS
Developer Tools: VScode, AutoCAD, Fusion360 , Android Studio, Jetbrains, Arduino IDE, Microchip Studio, RPi OS, Blynk IoT, RemoteXY.
Databases:SQL, MongoDB
Technical Expertise : Robotics, Electronics, ROS2 (Robot Operating System 2) , Machine Learning, Internet of Things(IoT), Control Systems , Embedded Systems , Sensor Integration , Mechanical Design , Web Development , App Development
Additional Projects: Quad copter , CNC Machine , Buglary Alarm System , Remote Controlled Car
Soft Skills: Effective communication , Leadership , Time management , Teamwork , Creativity, Problem Solving , Adaptability
Coursework: Electronic Engineering, Digital Electronics, AutoCAD, Process Instrumentation, Problem Solving and computer programming, MATLAB
Areas of Interest: Industrial and Mobile Robotics

Positions of Responsibility

•Joint Secretary, Robotics Club *2022-Present*

Achievements

•First Prize ECE's Hackathon NITW *November 7th 2023*

•4th Prize MJ College Hackathon *December 11th 2023*